**5. Missionaries and Cannibals Problem**

**Aim:**

To solve the Missionaries and Cannibals problem using Python.

**Algorithm:**

1. Start with 3 missionaries and 3 cannibals on the left bank.
2. Use a boat that can carry one or two people.
3. Ensure cannibals never outnumber missionaries on either bank when missionaries are present.
4. Generate valid boat crossings.
5. Repeat until everyone reaches the right bank.

**Program:**

```
from collections import deque
start = (3, 3, 0)
goal = (0, 0, 1)
moves = [(1, 0), (2, 0), (0, 1),
        (0, 2), (1, 1)]
def safe (m, c):
    return (0 <= m <= 3 and 0 <= c <= 3 and
            (m == 0 or m >= c) and
           (3-m == 0 or 3-m >= 3-c))
queue = deque ([(start, [])])
visited = {start}
while queue:
    state, path = queue.popleft()
    if state == goal:
        for step in path + [state]:
            print(step)
        break
    m, c, boat = state
```

```
    for dm, dc in moves:
        if boat == 0:
            new = (m-dm, c-dc, 1)
        else:
            new = (m+dm, c+dc, 0)
        if safe (new [0], new [1]) and new not in visited:
            visited.add(new)
            queue. append ((new, path + [state]))
```

**Output:**

```
==========
(3, 3, 0)
(3, 1, 1)
(3, 2, 0)
(3, 0, 1)
(3, 1, 0)
(1, 1, 1)
(2, 2, 0)
(0, 2, 1)
(0, 3, 0)
(0, 1, 1)
(1, 1, 0)
(0, 0, 1)
```

**Result:**

Thus, the Missionaries and Cannibals problem was solved using BFS.